MM-141
Materials Lab I

X-RAY DIFFRACTION

Experiment: 11

- Engr. Hamza Mahmood

Introduction

Motivation:

✓ X-ray diffraction is used to obtain structural information about crystalline solids.

- Measure the average spacings between layers or rows of atoms
- Determine the orientation of a single crystal or grain
- Find the crystal structure of an unknown material
- Measure the size, shape and internal stress of small crystalline regions

✓ Useful in biochemistry to solve the 3D structures of complex biomolecules.

✓ Bridge the gaps between physics, chemistry, and biology.

Background

1895 X-rays discovered by Roentgen

1914 First diffraction pattern of a crystal made by Knipping and von Laue

1915 Theory to determine crystal structure from diffraction pattern developed by Bragg

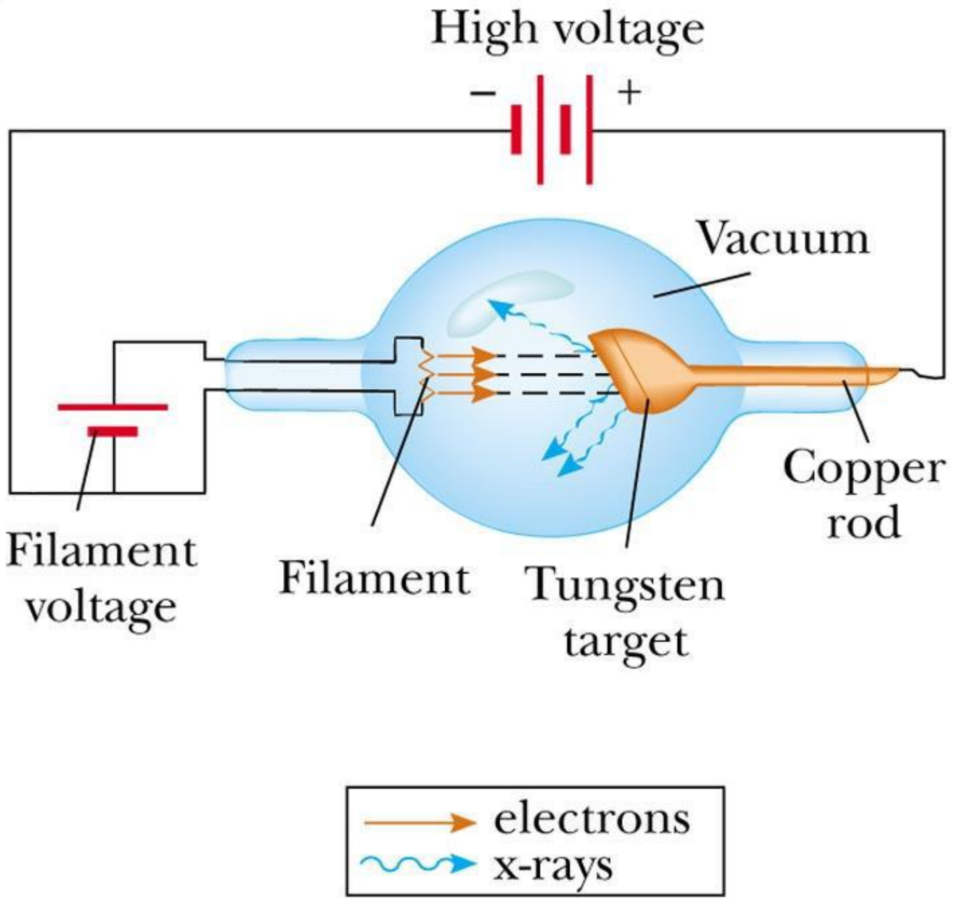
1953 DNA structure solved by Watson and Crick

Now Diffraction improved by computer technology; methods used to determine atomic structures and in medical applications

The first X-ray

X-Ray Production

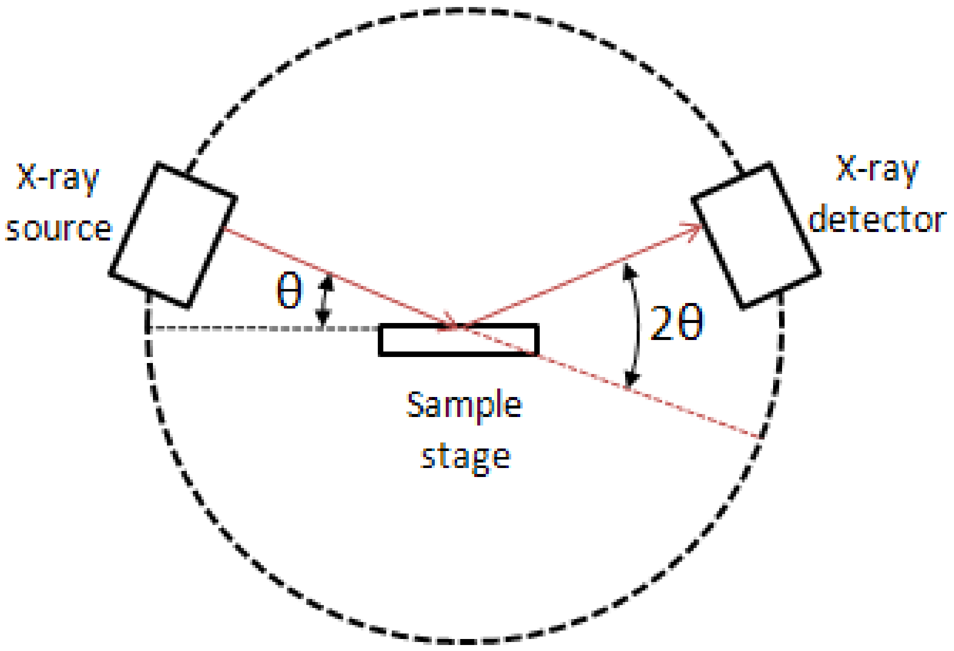
- ❑ When high energy electrons strike an anode in a sealed vacuum, x-rays are generated.
- ❑ Anodes are often made of copper, iron or molybdenum.
- ❑ X-rays are electromagnetic radiation of shorter wavelength and higher frequency

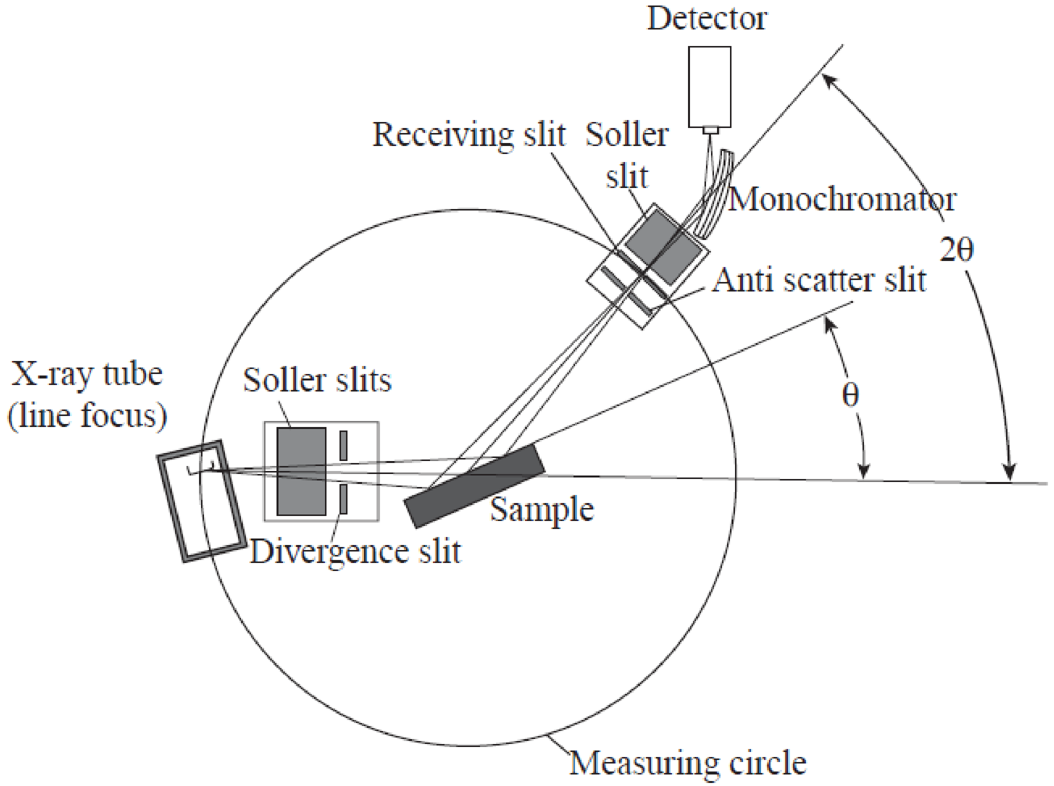


Production of X-Rays

X-ray Diffraction

- *A <u>scattering of X-rays</u> by the atoms of a crystal that produces an <u>interference effect</u> so that the diffraction pattern gives information on the <u>structure of the crystal</u> or the <u>identity</u> of a crystalline substance*





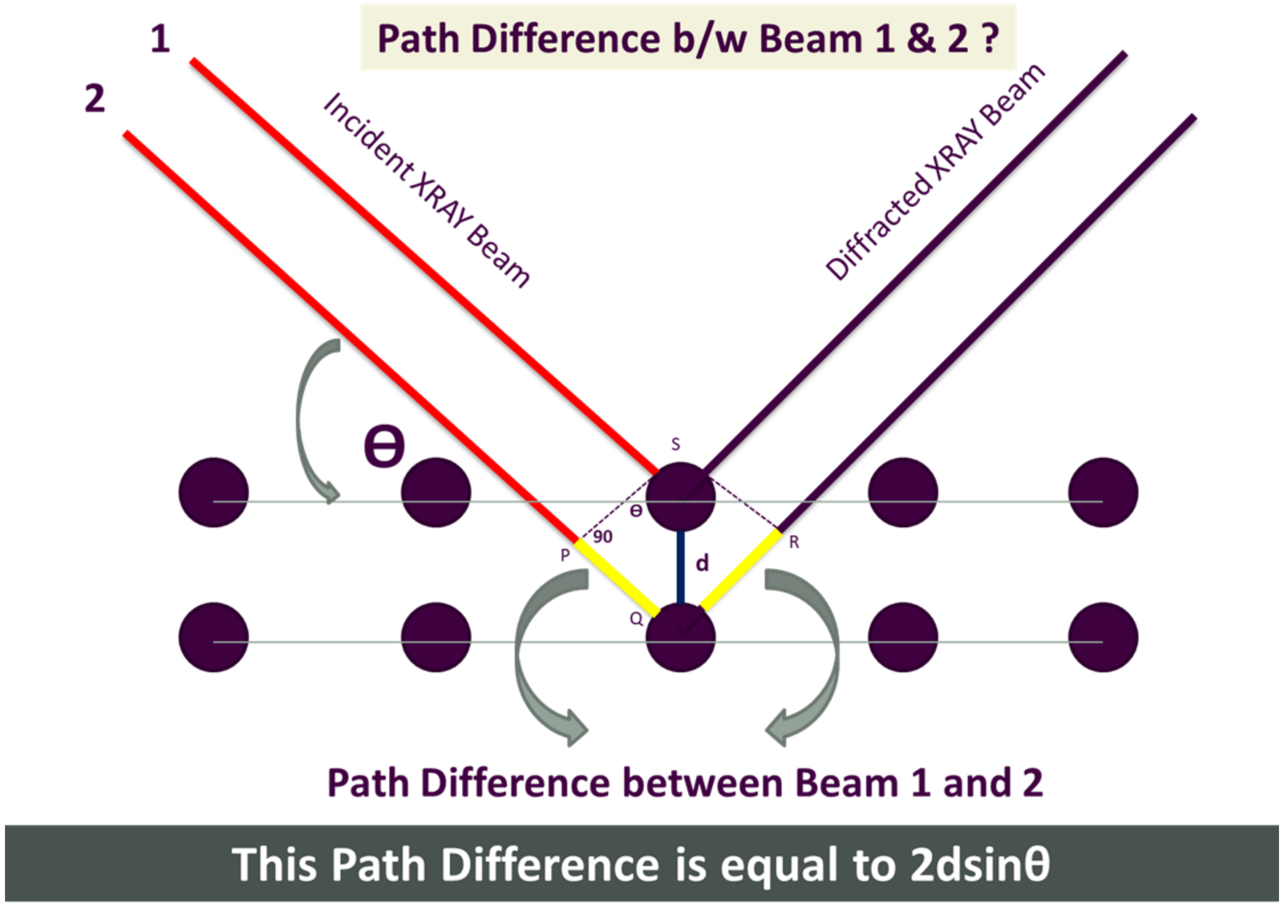
How Diffraction Works? Bragg's Law

- Bragg's Law **nλ=2dsinθ**

- The beam reflected from the lower surface travels farther than the one reflected from the upper surface

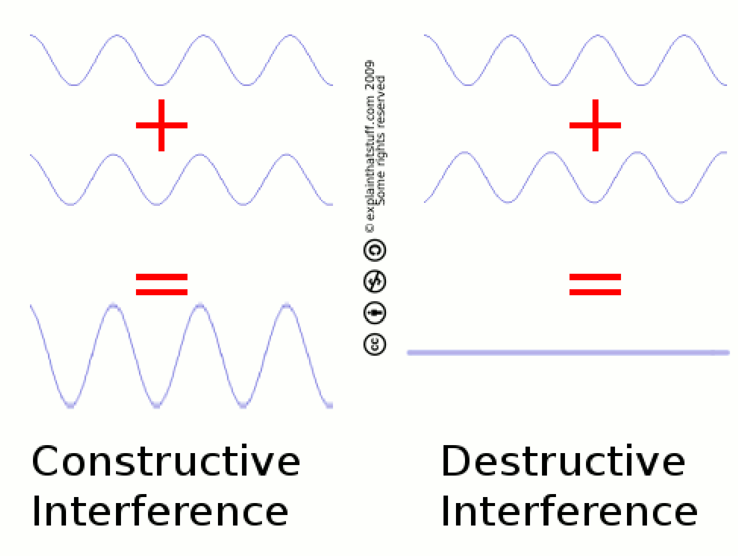
- The diffracted Beams *will be in phase* if the path difference is equal to multiple of wavelength, i.e. (nλ) constructive interference occurs

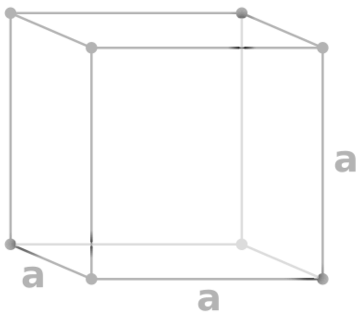
- Thus, Constructive Interference takes place if **nλ=2dsinθ** is satisfied. This is called Bragg's Diffraction

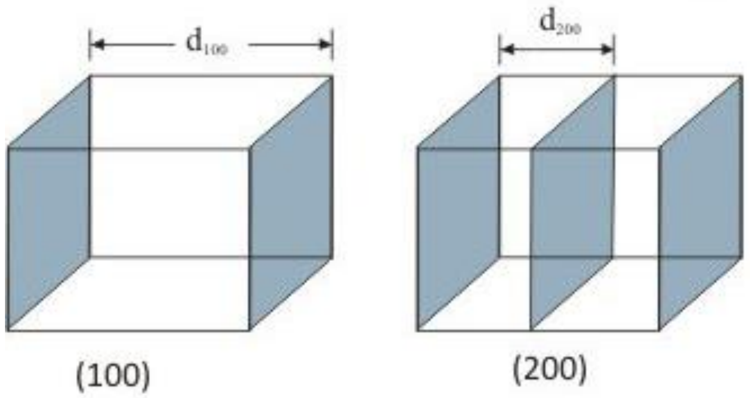


Terminology used in XRD

- **Diffraction**
  Scattering of X-Rays by atoms of a crystal
- **Constructive Versus Destructive Interference**
- **Lattice Parameter**
  Refers to the physical dimension of unit cells in a crystal lattice (a, b, and c)
- **Interplanar Spacing (d)**
  Distance between two planes (of the same type/family)
- **Phase**
  A homogeneous, physically distinct portion of matter (with a unique lattice parameter)



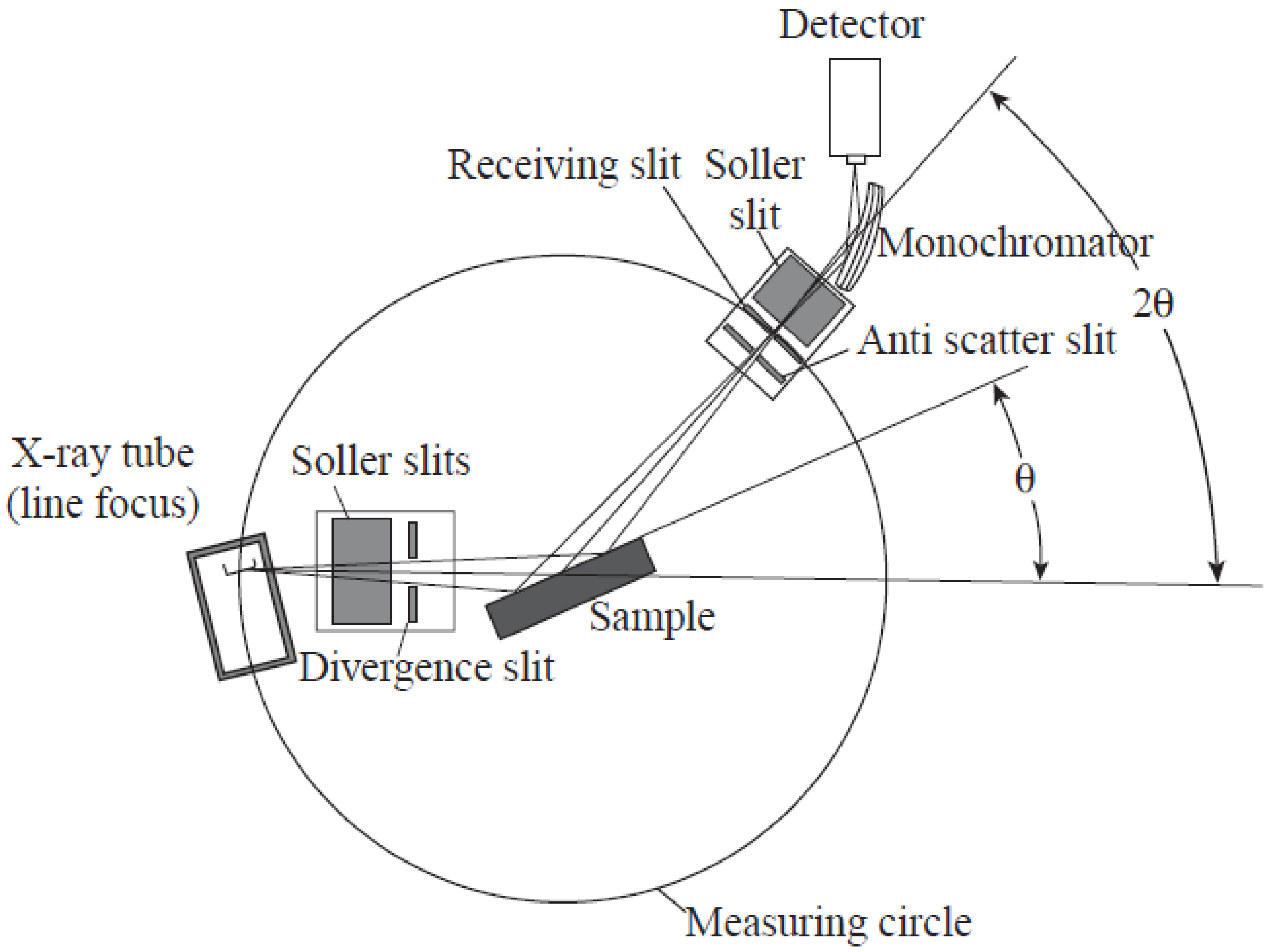


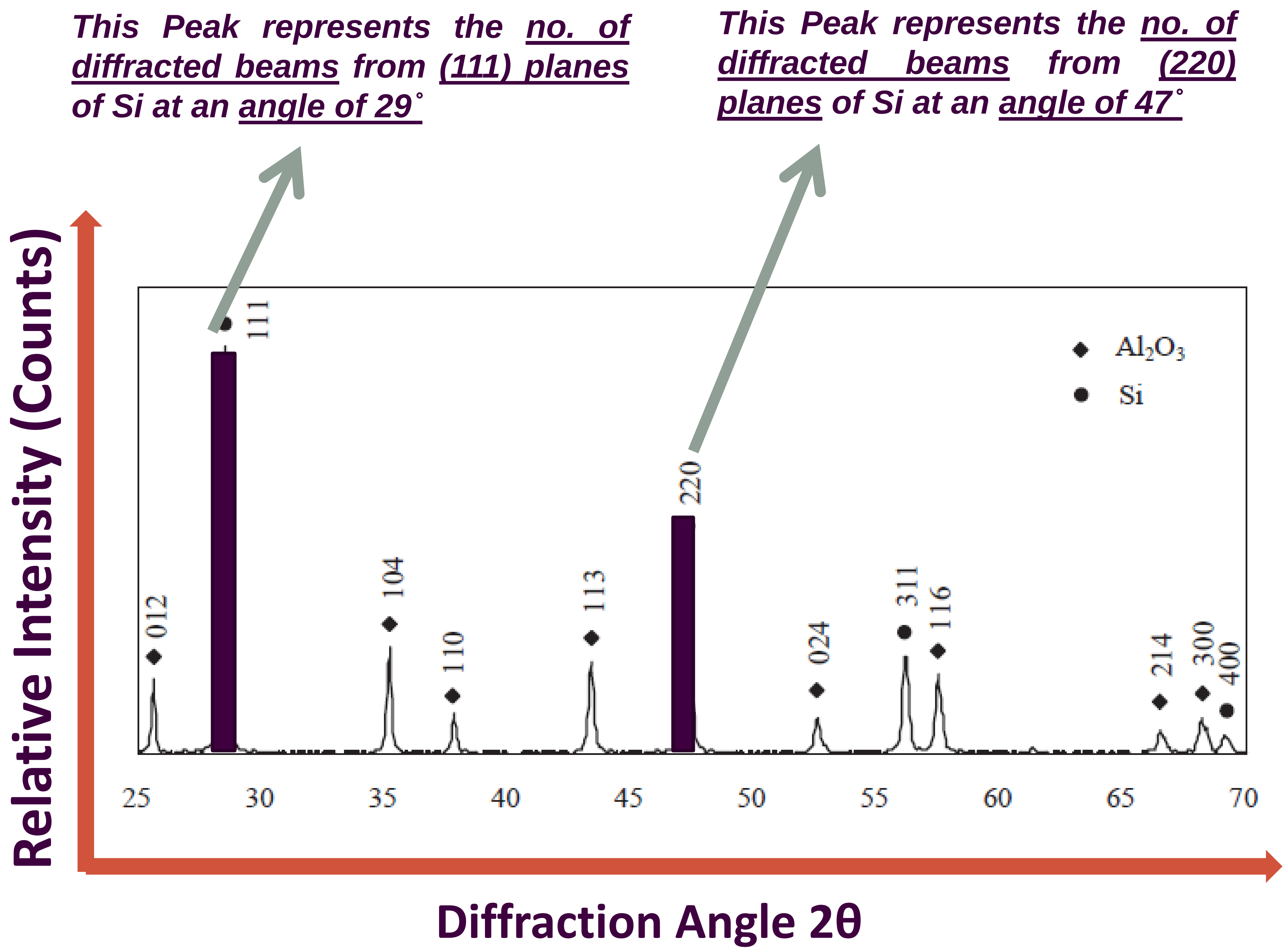


Bragg-Brentano Arrangement

To enable scanning over a range of ϴ

The X-ray incident beam is fixed, but a sample stage rotates at ϴ while detector rotates at 2ϴ




This Peak represents the no. of diffracted beams from (111) planes of Si at an angle of 29˚
This Peak represents the no. of diffracted beams from (220) planes of Si at an angle of 47˚
Relative Intensity (Counts)
Diffraction Angle 2θ
Al2O3
Si
012
111
104
110
113
220
024
311
116
214
300
400
25
30
35
40
45
50
55
60
65
70

Relationship between diffraction data and crystal parameters

For a cubic crystal system

$$n\lambda = 2d \sin\theta$$

Everything else except "d" is known... so d-spacing is found using the above equation

What about lattice parameter?

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

There are two unknown values in the second equation. Can we find lattice parameter from this Data?

Cohen's Least-Square Method or Extrapolation Method is used for determining Lattice Parameter.

The overall picture

Hit a sample by X-Rays

↓

The X-Rays that undergo Bragg's Diffraction are detected

↓

The Detected signals carry important Information about the material (Plane index and angle of Diffraction)

↓

The data are compared with previous information in the database using a software Xpert Highscore (having millions of diffraction spectra) for Phase Identification

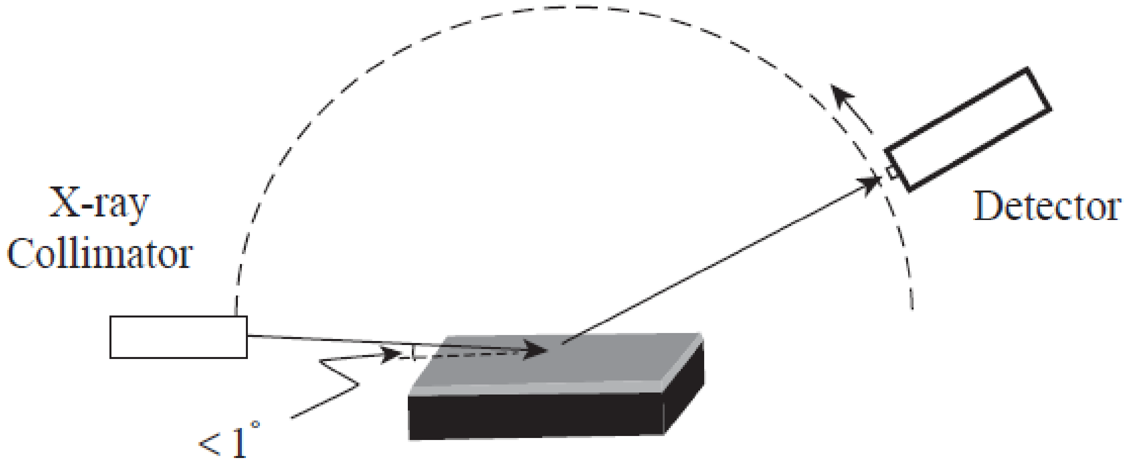
Thin film X-ray Diffractometry

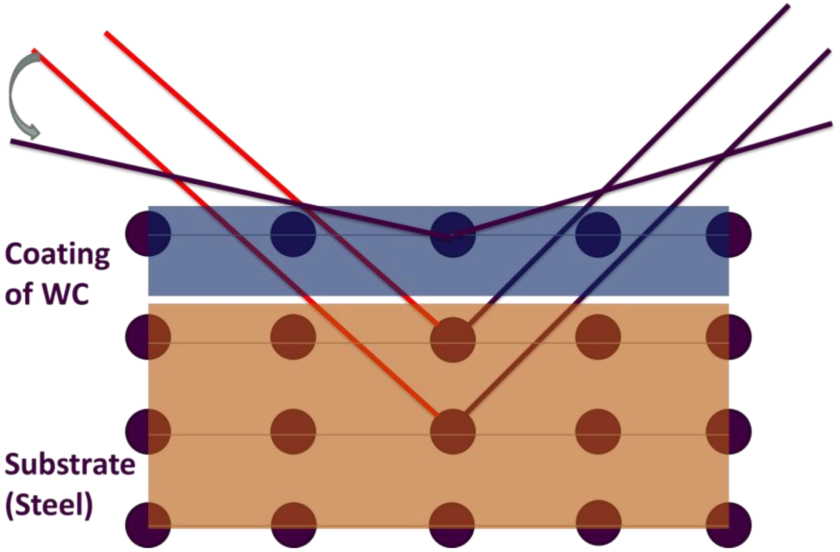
Detecting crystal structure of coatings

Signals are coming from substrate, but we want coating to be analyzed? What to Do?

Let us check what happens by decreasing incident angle of x-rays

Grazing incidence XRD (GIXRD) is often used for such samples, where the incident X-ray angle is very shallow to increase the interaction volume within the thin layer without penetrating too deeply into the substrate.





Applications of XRD

- ***Phase Identification***
- ***Quantitative Measurement:*** **Relative amounts of compounds or phases in a sample can be determined**
- ***Residual stresses***
- ***Preferred Orientation (Texture)***
- ***Studying High Temperature Phenomena taking place in a material like Phase transformation, grain growth***

References

[1] B. S. Saini and R. Kaur, “X-ray diffraction,” Handbook of Modern Coating Technologies. Elsevier, pp. 85–141, 2021. doi: 10.1016/b978-0-444-63239-5.00003-2.

[2] **Hacettepe Üniversitesi: X-Ray Diffraction (XRD)**
https://yunus.hacettepe.edu.tr/~selis/teaching/WEBkmu396/ppt/Presentations2010/XRD.pdf

[3] **The University of Oklahoma: X-Ray Diffraction**
https://web.pdx.edu/~pmoeck/phy381/Topic5a-XRD.pdf

Thank You for Your Attention!

Any Questions?